

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0614	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Y. Veera Reddy	Reg. No.:	192211898
Section / Batch:	A	Date:	
Instructions to Students • Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks. • Clearly show all steps in derivations and complexity analysis. • Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required. • Justify each step in recurrence solving using appropriate methods. • Neat presentation and logical explanation are mandatory			
Question Type		Focus Area	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Q1. A video streaming platform processes user data using an algorithm with recurrence $T(n) = 2T(n/2) + n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how this affects scalability as users increase. Interpret efficiency using asymptotic notation.

Q2. A social media platform processes posts using a recursive algorithm $T(n) = T(n-1) + n$. Apply the substitution method to solve the recurrence. Expand step-by-step and derive the final complexity. Analyze how performance degrades with large inputs. Interpret the result using asymptotic notation.

Code of Conduct

I (Y. Veera Reddy/192211898) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Veera Reddy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						
Faculty In-charge		Course Coordinator		Head of Department		
Signature: _____		Signature: _____		Signature: _____		
Date: _____		Date: _____		Date: _____		

Q1: Video Streaming Platform

ThGiven Recurrence

$$T(n) = 2T(n/2) + n$$

Step 1: Identify Master Theorem Parameters

The standard form of the Master Theorem is:

$$T(n) = aT(n/b) + f(n)$$

Here,

- $a = 2$
- $b = 2$
- $f(n) = n$

Step 2: Calculate $n \log_b a$

$$n \log_2 2 = n \cdot 1 = n$$

us,

- $f(n) = n$
- $n \log_b a = n$

Since both are equal,

$$f(n) = \Theta(n \log_b a)$$

Step 3: Apply Master Theorem

This satisfies Case 2 of the Master Theorem:

If

$$f(n) = \Theta(n \log_b a \log^k n)$$

where $k = 0$, then

$$T(n) = \Theta(n \log_b a \log^{k+1} n)$$

Therefore,

$$T(n) = \Theta(n \log n) \quad \boxed{\{T(n) = \Theta(n \log n)\}} \quad T(n) = \Theta(n \log n)$$

Scalability Analysis

- The algorithm divides the problem into two equal halves.
- Each level performs $O(n)$ work.
- There are $\log n$ levels.

As the number of users increases, execution time grows slightly faster than linear due to the logarithmic factor. This makes the algorithm highly scalable and suitable for large datasets.

Asymptotic Interpretation

$$\Theta(n \log n) \quad \boxed{\{ \Theta(n \log n) \}} \quad \Theta(n \log n)$$

This means the running time grows proportionally to $n \log n$, which is efficient for large-scale applications

Q2: Social Media Platform

Given Recurrence

$$T(n) = T(n-1) + n$$

Step 1: Expand Using Substitution

Expand repeatedly:

$$T(n) = T(n-1) + n$$

Substitute for $T(n-1)$:

$$= T(n-2) + (n-1) + n$$

Substitute again:

$$= T(n-3) + (n-2) + (n-1) + n$$

Continue until the base case:

$$= T(1) + 2 + 3 + 4 + \dots + n$$

Assume

$$T(1) = c$$

Hence,

$$T(n) = c + \sum_{i=2}^n i$$

Step 2: Evaluate the Sum

Using the formula,

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Therefore,

$$\sum_{i=2}^n i = \frac{n(n+1)}{2} - 1$$

Substitute:

$$T(n) = c + n(n+1) - 1 \quad T(n) = c + \frac{n(n+1)}{2} - 1 \quad T(n) = c + 2n(n+1) - 1$$

Simplify:

$$T(n) = n^2 + n + c - 1 \quad T(n) = \frac{n^2 + n}{2} + c - 1 \quad T(n) = 2n^2 + n + c - 1$$

The dominant term is

$$n^2 \quad \frac{n^2}{2} \quad 2n^2$$

Hence,

$$T(n) = \Theta(n^2) \quad \boxed{T(n) = \Theta(n^2)} \quad T(n) = \Theta(n^2)$$

Performance Analysis

- Each recursive call reduces the problem size by only 1.
- The work done increases linearly at every step.
- The total work becomes the sum of the first n integers, resulting in quadratic growth.

As the number of posts becomes very large, execution time increases rapidly, making the algorithm less scalable than divide-and-conquer algorithms.

Asymptotic Interpretation

$$\Theta(n^2) \quad \boxed{\Theta(n^2)} \quad \Theta(n^2)$$

This means the running time grows proportionally to the square of the input size, leading to significant performance degradation for large datasets.